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Exp.3 Scalling VLANs (DTP, VTP, STP)

Objectives

1. Illustrates the negotiation options of DTP.
2. Explain the need for VTP and its configuration.
3. Explain the role of redundancy in a converged network.
4. Summarize how STP works to eliminate Layer 2 loops in a converged network.

Dynamic Trunking Protocol (DTP)

Some Cisco switches have a proprietary protocol that lets them automatically negotiate trunking with a neighboring device. This protocol is called Dynamic Trunking Protocol (DTP). DTP can speed up the configuration process for a network administrator. Ethernet trunk interfaces support different trunking modes. An interface can be set to trunking or nontrunking, or to negotiate trunking with the neighbor interface. Trunk negotiation is managed by DTP, which operates on a point-to-point basis only, between network devices.

DTP is a Cisco proprietary protocol that is automatically enabled on Catalyst 2960 and Catalyst 3650 Series switches. DTP manages trunk negotiation only if the port on the neighbor switch is configured in a trunk mode that supports DTP. Switches from other vendors do not support DTP. Caution: Some internetworking devices might forward DTP frames improperly, which can cause misconfigurations. To avoid this, turn off DTP on Cisco switch interfaces that are connected to devices that do not support DTP.

The default DTP configuration for Cisco Catalyst 2960 and 3650 switches is dynamic auto.

To enable trunking from a Cisco switch to a device that does not support DTP, use the *switchport mode trunk* and *switchport nonegotiate* interface configuration mode commands. This causes the interface to become a trunk, but it will not generate DTP frames.

`S1(config-if)# switchport mode trunk`

`S1(config-if)# switchport nonegotiate`

{ automatically
trunk لا يفاوض }
بغير DTP frames

To re-enable dynamic trunking protocol use the *switchport mode dynamic auto* command.

`S1(config-if)# switchport mode dynamic auto`

{ default }

The *switchport mode* command has additional options for negotiating the interface mode. The full command syntax is the following:

`Switch(config-if)# switchport mode { access | dynamic { auto | desirable } | trunk }`

The options are described in the table below.

Option	Description
access	- Puts the interface (access port) into permanent nontrunking mode and negotiates to convert the link into a nontrunk link. - The interface becomes a nontrunk interface, regardless of whether the neighboring interface is a trunk interface.
dynamic auto	- Makes the interface able to convert the link to a trunk link. - The interface becomes a trunk interface if the neighboring interface is set to trunk or desirable mode. - The default switchport mode for all Ethernet interfaces is dynamic auto. <i>default status</i> <i>* access</i>
dynamic desirable	- Makes the interface actively attempt to convert the link to a trunk link. - The interface becomes a trunk interface if the neighboring interface is set to trunk, desirable, or dynamic auto mode. <i>trunk only if</i>
trunk	- Puts the interface into permanent trunking mode and negotiates to convert the neighboring link into a trunk link. - The interface becomes a trunk interface even if the neighboring interface is not a trunk interface.

The table below illustrates the results of the DTP configuration options on opposite ends of a trunk link connected to Catalyst 2960 switch ports. Best practice is to configure trunk links statically whenever possible.

	Dynamic Auto	Dynamic Desirable	Trunk	Access
Dynamic Auto	Access	Trunk	Trunk	Access
Dynamic Desirable	Trunk	Trunk	Trunk	Access
Trunk	Trunk	Trunk	Trunk	Limited connectivity
Access	Access	Access	Limited connectivity	Access

To determine and verify the current DTP mode, issue the `show dtp interface` command.

VLAN Trunking Protocol (VTP)

To carry traffic of a VLAN, it must be first configured on the switch. Suppose, if the user wants to send a frame from source to destination and the shortest path between them contains 1000 switches. To process a frame of any VLAN, VLANs should be configured first so, have to configure the same VLANs on all the 1000 switches manually. It will not be possible for the administrator to do that. Here comes VTP to the rescue.

VTP is CISCO proprietary protocol used to maintain consistency throughout the network or the user can say that synchronizing the VLAN information in the same VTP domain. VTP allows you

to add, delete and rename VLANs which is then propagated to other switches in the VTP domain. VTP advertisements can be sent over 802.1Q, and ISL trunks.

There are some requirements for VTP to communicate VLAN information between switches. These are:

- The VTP version must be same on the switches user wants to configure
- VTP domain name must be same on the switches
- One of the switches must be a server
- Authentication should match if applied

VTP modes – There are 3 modes:

- Server
- Client
- Transparent

Server – The switches are set to this mode by default. This mode allows you to create, add and delete VLANs. The changes you want to make should be done in this mode. Any changes that are done on this mode (on a particular switch) will be advertised to all the switches that are in the same VTP domain. In this mode, the configuration are saved in NVRAM.

Configuration:

Switch(config)#vtp mode server

Switch(config)#vtp domain ju

Switch(config)#vtp password cisco

User can verify the configuration by:

Switch(config)#do show vtp password

Switch(config)#do show vtp

Client – In this mode, the switches receive the updates and can also forward the updates to other switches (which are in the same VTP domain). The updates received here are not saved in NVRAM so all the configuration will be deleted if the switch is reset or reloaded i.e the switches will only learn and pass the VTP summary advertisements to the other switches.

Configuration:

Switch(config)#vtp mode client

Switch(config)#vtp domain ju

Switch(config)#vtp password cisco

Transparent – This mode only forwards the VTP summary advertisements through trunk link. The transparent mode switches can make their own local database which keep secret from other switches. The whole purpose of transparent mode is to forward the VTP summary advertisements but not to take part in the VLAN assignments.

Configuration:

Switch(config)#vtp mode transparent

Switch(config)#vtp domain ju

Switch(config)#vtp password cisco

Configuration Revision Number – The configuration revision number is a 32-bit number that indicates the level of revision for a VTP packet. This configuration number is tracked by every switch in order to find that the received information is more recent than the current version. Everytime one modification is done on the VLANs by the server switch, and the configuration revision number increases by one. The client mode devices receive it and check if the

* link should be trunk

default

* sh vtp status

* T

server ji updates jay

* T

* No add or remove vlan.

* Can add or remove vlan.

* T

+ -i ipi is
J delete
-vlan!

configuration revision number that they received is latest or not by comparing their own configuration number with the number received. If the configuration number is greater than their own number then the devices update their configuration and pass it to other clients of the same VTP domain. If the configuration number is the same then the devices just pass it to other clients of the same VTP domain. User can check the configuration revision number by:
`switch(config)#do show vtp status`

Spanning Tree Protocol (STP)

switching loop

{sh-spanning-tree}

Redundancy is an important part of the hierarchical design for eliminating single points of failure and preventing disruption of network services to users. Redundant networks require the addition of physical paths, but logical redundancy must also be part of the design. Having alternate physical paths for data to traverse the network makes it possible for users to access network resources, despite path disruption. However, redundant paths in a switched Ethernet network may cause both physical and logical Layer 2 loops.

Ethernet LANs require a loop-free topology with a single path between any two devices. A loop in an Ethernet LAN can cause continued propagation of Ethernet frames until a link is disrupted and breaks the loop. This Looping of frames causes three problems as stated below:

- MAC table instability – Due to looping of frame around LAN, MAC-Table of switch get changed frequently. Looping causes incorrect MAC-table entries resulting in incorrect frame delivery.
- Broadcast Storm – Repeated forwarding of frames around links in LAN causes the inefficient use of links.
- Multiple Frame Transmission – A very serious negative effect of looping is that multiple copies of same frame are delivered to host. This process left host with confusion.

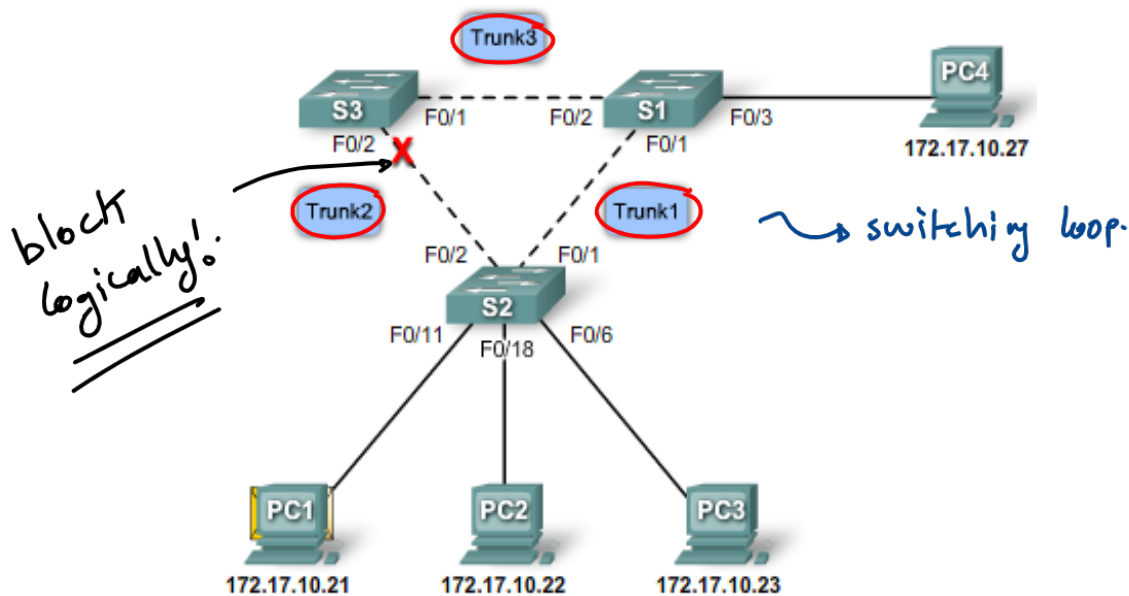
Unlike the Layer 3 protocols, IPv4 and IPv6, Layer 2 Ethernet does not include a mechanism to recognize and eliminate endlessly looping frames. Both IPv4 and IPv6 include a mechanism that limits the number of times a Layer 3 networking device can retransmit a packet. A router will decrement the TTL (Time to Live) in every IPv4 packet, and the Hop Limit field in every IPv6 packet. When these fields are decremented to 0, a router will drop the packet.

Spanning Tree Protocol (STP) is a loop-prevention network protocol that allows for redundancy while creating a loop-free Layer 2 topology. It ensures that there is only one logical path between all destinations on the network by intentionally blocking redundant paths that could cause a loop. A port is considered blocked when network traffic is prevented from entering or leaving that port. This does not include Bridge Protocol Data Unit (BPDU) frames that are used by STP to prevent loops. You will learn more about STP BPDU frames later in the experiment. Blocking the redundant paths is critical to preventing loops on the network. The physical paths still exist to provide redundancy, but these paths are disabled to prevent the loops from occurring. If the path is ever needed to compensate for a network cable or switch failure, STP recalculates the paths and unblocks the necessary ports to allow the redundant path to become active.

In the figure below, all switches now have STP enabled:

1. PC1 sends a broadcast out onto the network.
2. Switch S3 is configured with STP and has set the port for Trunk2 to a blocking state. The blocking state prevents ports from being used to forward switch traffic, preventing a loop from occurring. Switch S2 forwards a broadcast frame out all switch ports, except the originating port from PC1, and the port on Trunk2, which leads to the blocked port on S3.

3. Switch S1 receives the broadcast frame and forwards it out all of its switch ports, where it reaches PC4 and S3. S3 does not forward the frame back to S2 over Trunk2 because of the blocked port. The Layer 2 loop is prevented.



If the trunk link between switch S2 and switch S1 fails. Switch S3 unblocks the previously blocked port for Trunk2 and allows the broadcast traffic to traverse the alternate path around the network, permitting communication to continue. If this link comes back up, STP reconverges and the port on S3 is again blocked.

How STP Works?

STP uses the Spanning Tree Algorithm (STA) to determine which switch ports on a network need to be configured for blocking to prevent loops from occurring. The STA designates a single switch as the root bridge and uses it as the reference point for all path calculations. In the figure below the root bridge, switch S1, is chosen through an election process. All switches participating in STP exchange BPDUs to determine which switch has the lowest bridge ID (BID) on the network. The switch with the lowest BID automatically becomes the root bridge for the STA calculations.

The BPDU is the message frame exchanged by switches for STP. Each BPDU contains a BID that identifies the switch that sent the BPDU. The BID contains a priority value, the MAC address of the sending switch, and an optional extended system ID. The lowest BID value is determined by the combination of these three fields. Example of BID: {32768 - 0002.4A7C.4BC4}.

After the root bridge has been determined, the STA calculates the shortest path to the root bridge. Each switch uses the STA to determine which ports to block. While the STA determines the best paths to the root bridge for all destinations in the broadcast domain, all traffic is prevented from forwarding through the network. The STA considers both path and port costs when determining which path to leave unblocked. The path costs are calculated using port cost values associated with port speeds for each switch port along a given path. The sum of the port cost values determines the overall path cost to the root bridge. If there is more than one path to choose from, STA chooses the path with the lowest path cost.

When the STA has determined which paths are to be left available, it configures the switch ports into distinct port roles. The port roles describe their relation in the network to the root bridge and whether they are allowed to forward traffic.

Port Roles

There are four distinct port roles that switch ports are automatically configured for during the spanning-tree process.

Root Port - The root port exists on non-root bridges and is the switch port with the best path to the root bridge. Root ports forward traffic toward the root bridge. One root port is allowed per bridge. In the example below, switch S1 is the root bridge and switches S2 and S3 have root ports defined on the trunk links connecting back to S1.

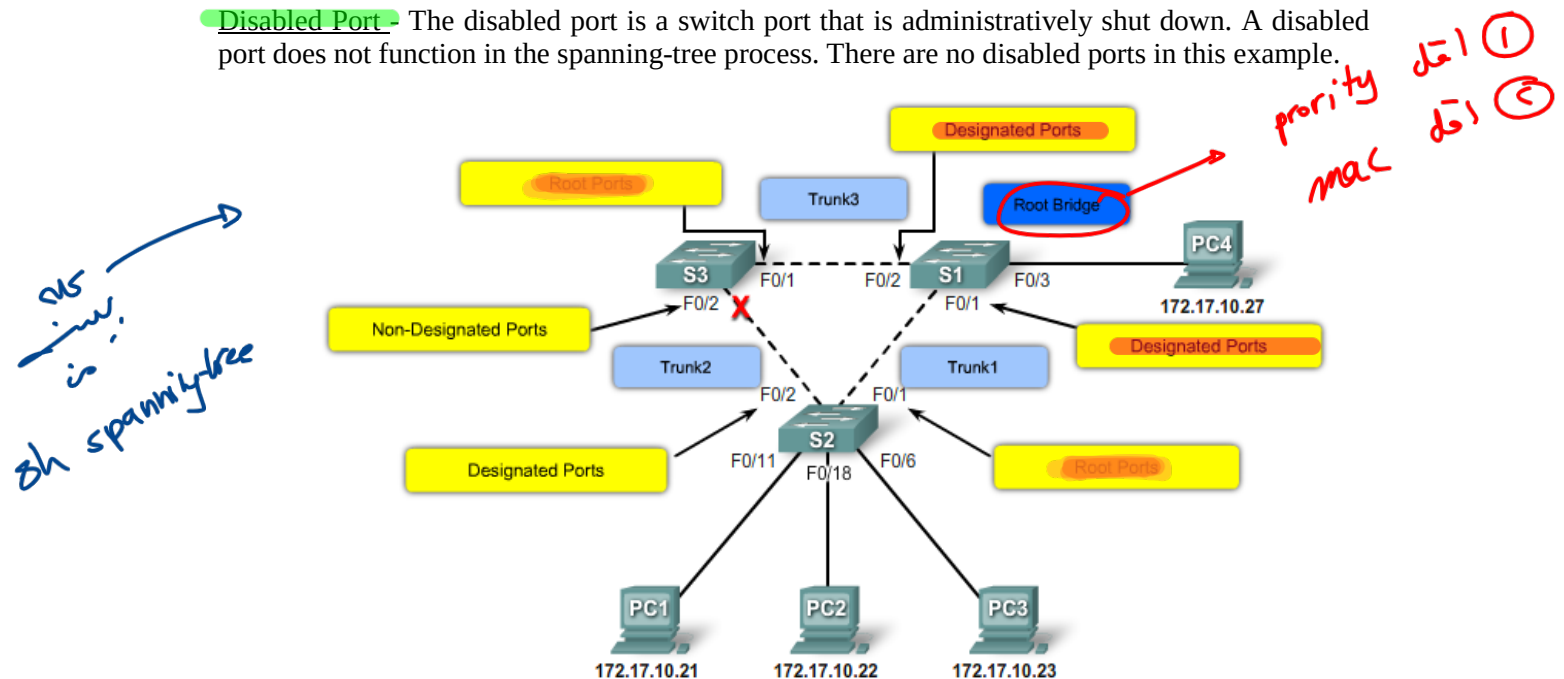
Designated Port - The designated port exists on root and non-root bridges. For root bridges, all switch ports are designated ports. For non-root bridges, a designated port is the switch port that receives and forwards frames toward the root bridge as needed. Only one designated port is allowed per segment. If multiple switches exist on the same segment, an election process determines the designated switch, and the corresponding switch port begins forwarding frames for the segment. Designated ports are capable of populating the MAC table.

In the example below, switch S1 has both sets of ports for its two trunk links configured as designated ports. Switch S2 also has a designated port configured on the trunk link going toward switch S3.

Non-designated Port - The non-designated port is a switch port that is blocked, so it is not forwarding data frames and not populating the MAC address table with source addresses. A non-designated port is not a root port or a designated port. For some variants of STP, the non-designated port is called an alternate port.

In the example below, switch S3 has the only non-designated ports in the topology. The non-designated ports prevent the loop from occurring.

Disabled Port - The disabled port is a switch port that is administratively shut down. A disabled port does not function in the spanning-tree process. There are no disabled ports in this example.



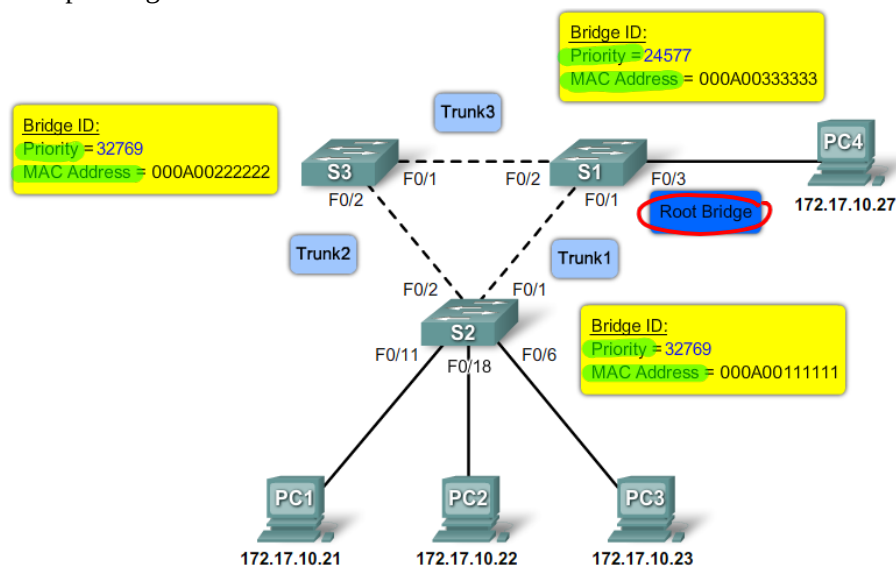
The Root Bridge

Every spanning-tree instance (switched LAN or broadcast domain) has a switch designated as the root bridge. The root bridge serves as a reference point for all spanning-tree calculations to determine which redundant paths to block.

An election process determines which switch becomes the root bridge. The figure below shows the BID fields. The BID is made up of a priority value, an extended system ID, and the MAC address of the switch.

All switches in the broadcast domain participate in the election process. After a switch boots, it sends out BPDU frames containing the switch BID and the root ID every 2 seconds. By default, the root ID matches the local BID for all switches on the network. The root ID identifies the root bridge on the network. Initially, each switch identifies itself as the root bridge after bootup.

As the switches forward their BPDU frames, adjacent switches in the broadcast domain read the root ID information from the BPDU frame. If the root ID from the BPDU received is lower than the root ID on the receiving switch, the receiving switch updates its root ID identifying the adjacent switch as the root bridge. Note: It may not be an adjacent switch, but any other switch in the broadcast domain. The switch then forwards new BPDU frames with the lower root ID to the other adjacent switches. Eventually, the switch with the lowest BID ends up being identified as the root bridge for the spanning-tree instance.



Best Paths to the Root Bridge

When the root bridge has been designated for the spanning-tree instance, the STA starts the process of determining the best paths to the root bridge from all destinations in the broadcast domain. The path information is determined by summing up the individual port costs along the path from the destination to the root bridge.

The default port costs are defined by the speed at which the port operates, 10-Gb/s Ethernet ports have a port cost of 2, 1-Gb/s Ethernet ports have a port cost of 4, 100-Mb/s Fast Ethernet ports have a port cost of 19, and 10-Mb/s Ethernet ports have a port cost of 100.

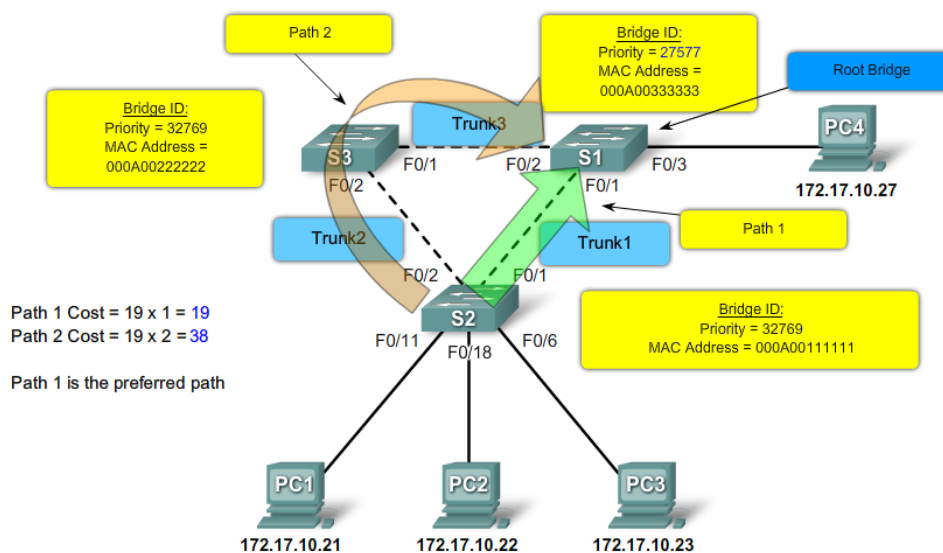
Although switch ports have a default port cost associated with them, the port cost is configurable as shown in the figure below. The ability to configure individual port costs gives the administrator the flexibility to control the spanning-tree paths to the root bridge.


```

S2#configure terminal
Enter configuration commands, one per line. End with CNTL/Z.
S2(config)#interface f0/1
S2(config-if)#spanning-tree cost 25
S2(config-if)#end
S2#

```

Path cost is the sum of all the port costs along the path to the root bridge. The paths with the lowest path cost become the preferred path, and all other redundant paths are blocked. In the example, the path cost from switch S2 to the root bridge switch S1, over path 1 is 19 (based on the IEEE-specified individual port cost), while the path cost over path 2 is 38. Because path 1 has a lower overall path cost to the root bridge, it is the preferred path. STP then configures the redundant path to be blocked, preventing a loop from occurring.



To verify the port and path cost to the root bridge, enter the show spanning-tree privileged EXEC mode command. The Cost field in the output is the total path cost to the root bridge. This value changes depending on how many switch ports need to be traversed to get to the root bridge. In the output, each interface is also identified with an individual port cost of 19.

```

S2#show spanning-tree

VLAN0001
  Spanning tree enabled protocol ieee
  Root ID    Priority 27577
             Address  000A.0033.3333
             Cost    19
             Port    1
             Hello Time 2 sec  Max Age 20 sec  Forward Delay 15 sec

  Bridge ID  Priority 32769 (priority 32768 sys-id-ext 1)
             Address  000A.0011.1111
             Hello Time 2 sec  Max Age 20 sec  Forward Delay 15 sec
             Aging Time 300

Interface Role Sts Cost Prio.Nbr Type
-----
F0/1      Root FWD 19    128.1   Edge P2p
F0/2      Desg FWD 19    128.2   Edge P2p

```


Configure and Verify the BID

When a specific switch is to become a root bridge, the bridge priority value needs to be adjusted to ensure it is lower than the bridge priority values of all the other switches on the network. There are two different configuration methods that you can use to configure the bridge priority value on a Cisco Catalyst switch.

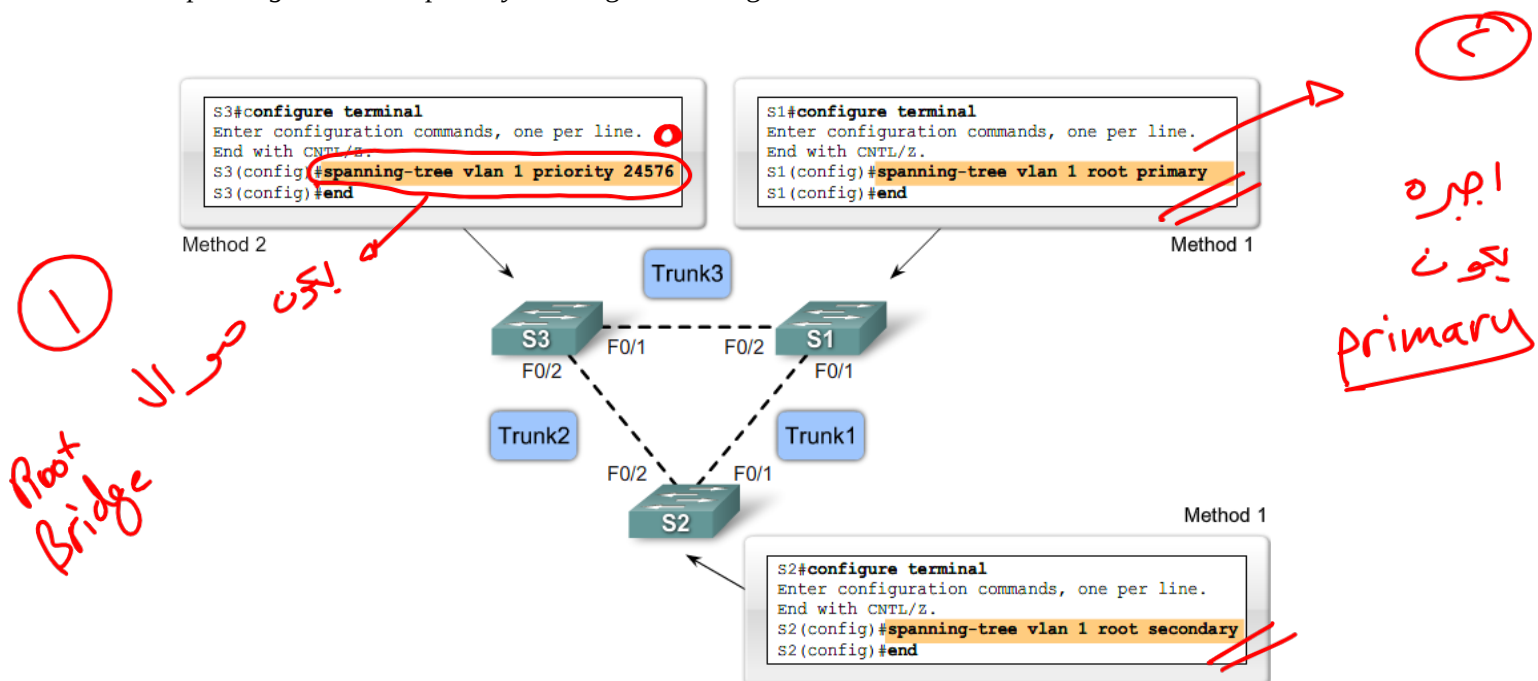
Method 1 - To ensure that the switch has the lowest bridge priority value, use the *spanning-tree vlan vlan-id root primary* command in global configuration mode. The priority for the switch is set to the predefined value of 24576 or to the next 4096 increment value below the lowest bridge priority detected on the network.

If an alternate root bridge is desired, use the *spanning-tree vlan vlan-id root secondary* global configuration mode command. This command sets the priority for the switch to the predefined value of 28672. This ensures that this switch becomes the root bridge if the primary root bridge fails and a new root bridge election occurs and assuming that the rest of the switches in the network have the default 32768 priority value defined.

In the example below, switch S1 has been assigned as the primary root bridge using the *spanning-tree vlan 1 root primary* global configuration mode command, and switch S2 has been configured as the secondary root bridge using the *spanning-tree vlan 1 root secondary* global configuration mode command.

Method 2 - Another method for configuring the bridge priority value is using the *spanning-tree vlan vlan-id priority value* global configuration mode command. This command gives you more granular control over the bridge priority value. The priority value is configured in increments of 4096 between 0 and 65536.

In the example below, switch S3 has been assigned a bridge priority value of 24576 using the *spanning-tree vlan 1 priority 24576* global configuration mode command.



To verify the bridge priority of a switch, use the *show spanning-tree* privileged EXEC mode command. In the example, the priority of the switch has been set to 24576. Also notice that the switch is designated as the root bridge for the spanning-tree instance.

```

S1#show spanning-tree
VLAN0001
  Spanning tree enabled protocol ieee
  Root ID    Priority    24577
             Address     000A.0033.3333
             This bridge is the root
             Hello Time  2 sec  Max Age 20 sec  Forward Delay 15 sec

  Bridge ID  Priority    24577 (priority 24576 sys-id-ext 1)
             Address     000A.0033.3333
             Hello Time  2 sec  Max Age 20 sec  Forward Delay 15 sec
             Aging Time  300

Interface        Role Sts Cost      Prio.Nbr Type
-----
Fa0/1            Desg FWD 4         128.1   Shr
Fa0/2            Desg FWD 4         128.2   Shr
S1#

```

Port States

STP determines the logical loop-free path throughout the broadcast domain. The spanning tree is determined through the information learned by the exchange of the BPDU frames between the interconnected switches. To facilitate the learning of the logical spanning tree, each switch port transitions through five possible port states and three BPDU timers.

The spanning tree is determined immediately after a switch is finished booting up. If a switch port were to transition directly from the blocking to the forwarding state, the port could temporarily create a data loop if the switch was not aware of all topology information at the time. For this reason, STP introduces five port states. The table below summarizes what each port state does. The following provides some additional information on how the port states ensure that no loops are created during the creation of the logical spanning tree.

Processes	Blocking	Listening	Learning	Forwarding	Disable
Receives and process BPDUs	YES	YES	YES	YES	NO
Forward data frames received on interface	NO	NO	NO	YES	NO
Forward data frames switched from another interface	NO	NO	NO	YES	NO
Learn MAC addresses	NO	NO	YES	YES	NO

Blocking - The port is a non-designated port and does not participate in frame forwarding. The port receives BPDU frames to determine the location and root ID of the root bridge switch and what port roles each switch port should assume in the final active STP topology.

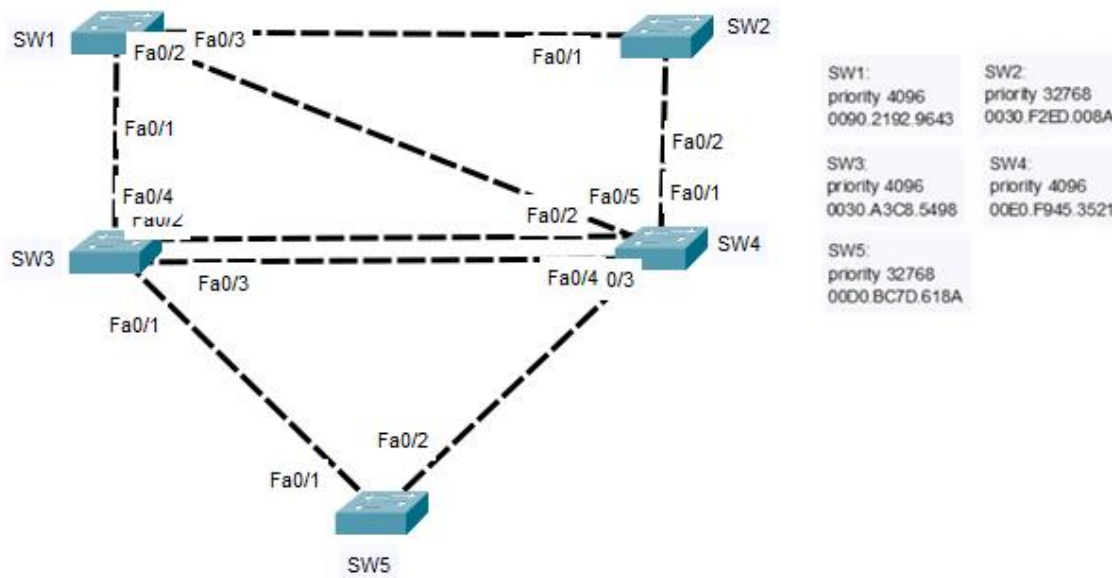
Listening - STP has determined that the port can participate in frame forwarding according to the BPDU frames that the switch has received thus far. At this point, the switch port is not only receiving BPDU frames, it is also transmitting its own BPDU frames and informing adjacent switches that the switch port is preparing to participate in the active topology.

Learning - The port prepares to participate in frame forwarding and begins to populate the MAC address table.

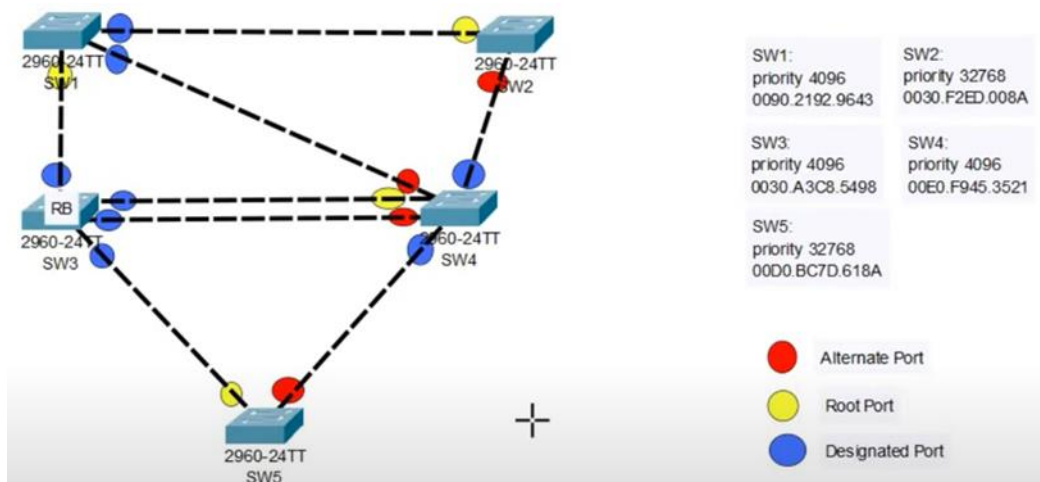
Forwarding - The port is considered part of the active topology and forwards frames and also sends and receives BPDU frames.

Disabled - The Layer 2 port does not participate in spanning tree and does not forward frames. The disabled state is set when the switch port is administratively disabled.

As a summary on how STP Works, consider the following topology and determine the port state of all interfaces:



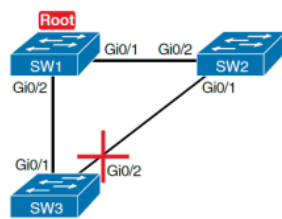
- 1- Elect a root bridge: The switch with the lowest BID (1- Priority, 2- MAC Address)
- 2- Elect a root port for each switch except the RB: The port that has the lowest cost path to the root bridge (Highest Bandwidth & Lowest number of links)
 - If a switch has multiple equal cost paths to the RB, it will select the path with the lowest neighbour's BID (If it is connected directly to the RB or has multiple links connected to that neighbour, it will select the lowest Port ID (Example of Port ID: Fa0/1))
- 3- All the ports that are opposite the root ports and all the root bridge's ports will be designated ports.
- 4- Elect a designated port for each remaining link: The switch that has the lowest cost path to the root bridge
 - If both switches have the same cost path, it will select the switch with lowest BID.
- 5- All the other ports will be alternate ports (Blocking state)



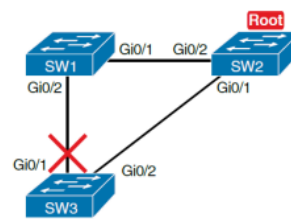
MSTP: Multiple Spanning Tree Protocol

- Developed by Cisco and IEEE, defined in 802.1S
- Fast
- Provides a spanning tree instance for group of VLANs

Instance 1: VLAN 100 - 120



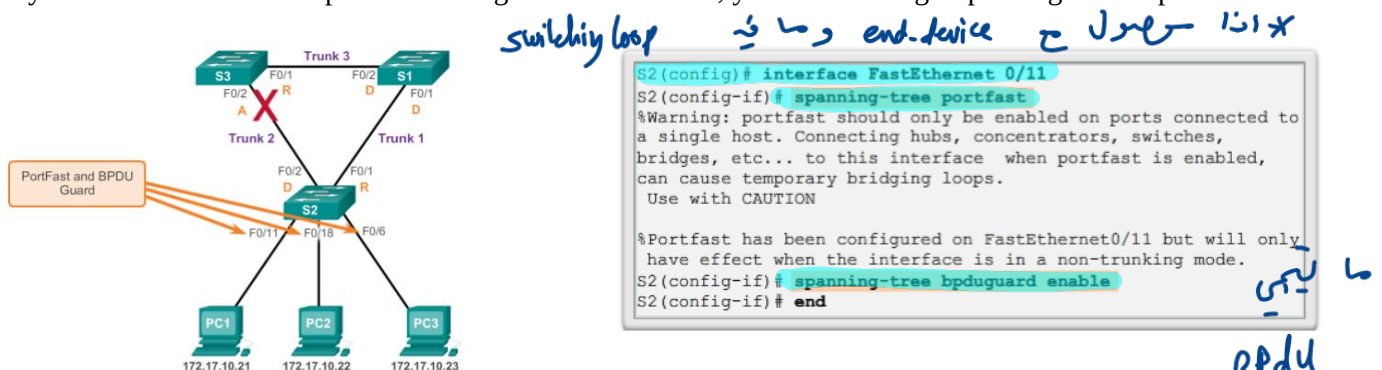
Instance 2: VLAN 121 - 140



PortFast and BPDU Guard

When a device is connected to a switch port or when a switch powers up, the switch port goes through both the listening and learning states, each time waiting for the Forward Delay timer to expire. This delay is 15 seconds for each state, listening and learning, for a total of 30 seconds. This delay can present a problem for DHCP clients trying to discover a DHCP server. DHCP messages from the connected host will not be forwarded for the 30 seconds of Forward Delay timers and the DHCP process may timeout. The result is that an IPv4 client will not receive a valid IPv4 address.

When a switch port is configured with PortFast, that port transitions from blocking to forwarding state immediately, bypassing the usual 802.1D STP transition states (the listening and learning states) and avoiding a 30 second delay. Because the purpose of PortFast is to minimize the time that access ports must wait for spanning tree to converge, it should only be used on access ports. If you enable PortFast on a port connecting to another switch, you risk creating a spanning tree loop.



In a valid PortFast configuration, BPDUs should never be received on PortFast-enabled switch ports because that would indicate that another bridge or switch is connected to the port. This potentially causes a spanning tree loop. To prevent this type of scenario from occurring, Cisco switches support a feature called BPDU guard. When enabled, BPDU guard immediately puts the switch port in an errdisabled (error-disabled) state on receipt of any BPDU. This protects against potential loops by effectively shutting down the port.

Procedures:

Dear students, please note that the lab problems **sheet**, the **packet tracer activities** and the **practical discussion videos** have been uploaded on your Microsoft Teams group. You are required to carefully study this experiment and then complete the lab sheet.

References

Cisco Networking Academy - CCNA: Switching, Routing, and Wireless Essentials.
<https://www.netacad.com>